

Setting up Your Databricks Environment



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Overview



Set up the workspace

Create a cluster

Work with a notebook

Configure the security

Walk through the scenario



Setting up Workspace



Workspace

Isolation Unit

Each workspace is isolated from others

Workspace ID

Each workspace has an identifier

Locked Resources

Deployed in control plane and data plane

Organize Assets

Notebooks, Libraries, Dashboards etc.

Access Control

Define access control on all assets



Creating Cluster



Cluster Types

Interactive Cluster

Interactively analyze the data

Created by users

Manually terminate

Option to auto terminate, if inactive

Low execution time

Auto scale on demand

Comparatively costly

Job Cluster

Run automated jobs

Auto created when job starts

Terminates when the job ends

Option to auto terminate not applicable

High throughput

Auto scale on demand

Comparatively cheaper



Cluster Modes

Standard Mode

Single user

No fault isolation

No task preemption

Each user require separate cluster

Supports Scala, Python, SQL, R & Java

High Concurrency Mode

Multiple users

Fault isolation

Task preemption – fair resource sharing

Maximum cluster utilization

Only supports Python, SQL & R



Pool



Cluster 1

Cluster 2



Pool Properties

Idle Instance Auto Termination

Minimum Idle Instances

Maximum Capacity

Instance Type



Working with Notebook



Configuring Security



Security Layers

Infrastructure

Workspace security using
Control plane & Data plane
VNETs, security groups,
TLS and resource locking

Identity

User authentication to
workspace using Azure
Active Directory

Asset

Built-in support for fine-
grain user permissions to
clusters, folders,
notebooks, jobs & data



Scenario Walkthrough





Attributes

- Pickup & drop time
- Pickup & drop location details
- Trip distance
- Passenger count
- Solo / shared ride
- Payment – Tips, tolls, tax etc.
- Payment mode



Capture data once a month (CSV files)

Extract on-prem using ETL tool

Build dimensions / facts

Store in RDBMS (data warehouse)

Build aggregated reports & KPIs

- Revenue by taxi type, location etc.
- Total trips, max trips by region etc.





For Hire Vehicles (FHV)

- Community Cars, Black Cars, Luxury Limousines, High Volume For-Hire Services

Dispatched from a base

Operate in different ways

Exponential increase in data volumes

Different schemas & formats

Merging & comparison of all taxi data

Granular & new aggregated reports

Faster turnaround time

Raw data for analysis





Build a Data Lake

Capture raw data in Data Lake

Use Azure Databricks to profile, merge & process data

Build dimensions, facts, KPIs & reports

Store data back to Data Lake

Allow analysts & data scientists to consume data



Summary



Organize Databricks assets in workspace

Create an interactive cluster

Use pools to speed up cluster start and scale times

Create and attach a notebook to cluster

- Autocomplete, Revision history, Git integration, Comments and Magic commands

Configure security for all assets

Taxi Service scenario

